

A 0.28-in
(7-mm) thick
wetsuit can keep
a diver alive in
water as cold
as 50°F (10°C).

STAYING WARM IN THE WATER

WETSUITS

Even on a hot day, water can steal enough heat from the human body to give a person hypothermia (a dangerous drop in body temperature). Unlike marine animals such as whales, which have a layer of fat to stay warm, humans need artificial coverings such as wetsuits to stay warm in cold water. Wetsuits let water in through the neck, sleeves, and legs, but very little heat can escape through the material, so the thin layer of water and the body keep warmer for longer.



FLUFFY COAT

Animals from cold, harsh environments are equipped with their own insulation. Polar bears rely on their thick fur and fat to survive in the freezing cold of the Arctic, with the thick layers providing an insulating effect similar to that of a wetsuit.

INSULATING MATERIALS

The secret to a wetsuit's insulating properties is a layer of material called neoprene. Neoprene is a synthetic (human-made) form of rubber made into a foam with many bubbles of air inside, which slow the loss of heat. A thin layer of water also gets trapped between the suit and body and is warmed up by a diver's body heat. Together, the layer of warm water and the neoprene layer help keep the diver warm in the cold water.

